

Short Questions oil & fats

① Write down the major & minor components of fats & oils

Fats & oils are also known as lipids are composed of major & minor components.

Major Component

Triglycerides

Fatty acids

Glycerol

Minor Component

Phospholipids

Sterols

Vitamins

These major & minor components collectively contribute to the nutritional value, flavour, texture and stability of fats & oils in foods and in our bodies.

2) What is emulsion? Give its types

Emulsions are a type of colloidal dispersion where two immiscible liquids, typically oil & water are mixed together with the help of an emulsifying agent which stabilize the mixture by preventing the liq. from separating into distinct layers.

4- Types of emulsion:-

- oil in water (O/W)
- water in oil (W/O)
- Multiple or complex emulsion
- Micro emulsion

③ Describe the reason of fat spoilage:

Fat spoilage is primarily caused by oxidation which occurs when fat is exposed to oxygen. Exposure of light, esp. UV light can break down fats and make them rancid. Additionally, high temp., moisture, micro-organisms and contaminants can contribute to spoilage.

④ Enlist the name and sources of 4 veg. oils.

Olive oil: From olives, primarily grown in Mediterranean countries.

Sunflower oil: From sunflower seeds, widely cultivated worldwide.

Canola oil: Made from rapeseed, commonly grown in Canada & US.

Coconut oil: Extracted from meat of coconut, commonly found in tropical regions.

⑤ Diff. b/w simple & mixed triglycerides

Simple

• Consist of 3 identical fatty acids molecule attached to glycerol backbone.

Mixed:-

• Consist of 2 or 3 diff. type of fatty acids molecule attached to glycerol backbone.

- They are relatively rare in nature not as commonly found in dietary fats.

- They are found in small quantities in some animal fats and certain plant oil.

e.g.: Stearin, Triolein

- They are the most common form of triglycerides and are found in majority of dietary fats & oils.

- The diff. combination of fatty acids in mixed triglycerides contribute to diverse physical properties & health effects of various dietary fats.

e.g.: Canola oil, Soybean oil.

⑥ Dist. b/w primary petrochemicals & intermediate petrochemicals. Give example.

Primary

They are basic building blocks derived directly from crude oil or natural gas.

Example :-

Ethylene: Used to produce plastics like polyethylene.

Intermediate

They are produced by further processing primary petrochemicals they serve as intermediate for various chemical products.

Ethylene oxide: Used in production of detergents and antifreeze.

Propylene: Used in production of plastics, synthetic rubber and chemicals.

Methanol: Used in manufacture of formaldehyde, acetic acid & various chemicals.

Acrylonitrile: Used in the production of acrylic fibre and plastics.

Ethylene glycol: Used as a coolant in automobiles and for making polyester fibers.

⑦ Role of catechol and pyrogallol in veg. tanning.

Catechol and pyrogallol are both polyphenolic compounds commonly used in veg. tanning process. Catechol and pyrogallol tannins are capable of cross-linking collagen fibers within the hide. This cross-linking imparts strength and durability to leather.

Both are important tannins in veg. tanning due to their ability to bind with collagen, provide color and increase water resistance. These properties make them valuable in production of high quality leather goods.

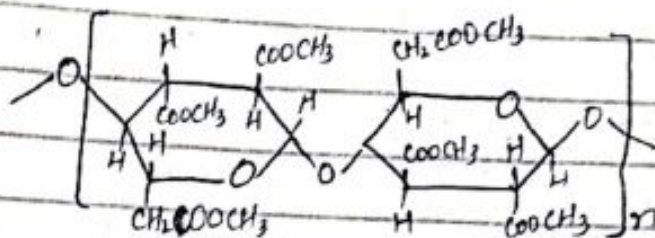
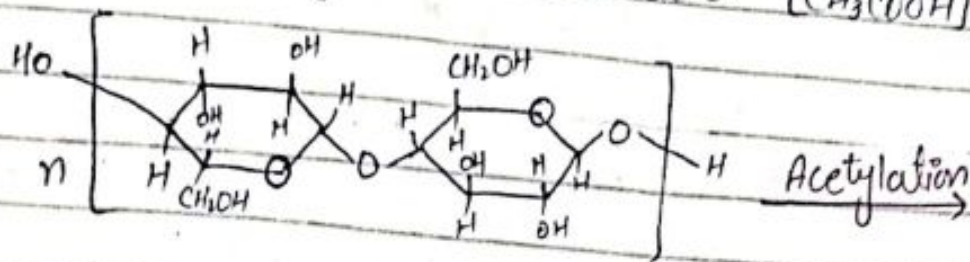
⑧ Write down the chemical reaction involved in manufacturing of cellulose acetate.

STEP # 1:-

Purification of cotton linters or wood pulp.

STEP #2:-

Purified cotton linters or wood pulps are steeped in glacial acetic acid. $[CH_3COOH]$.

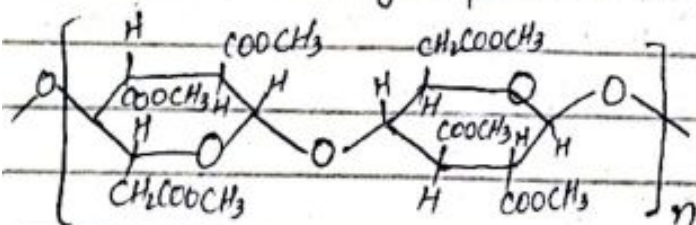


STEP #3:-

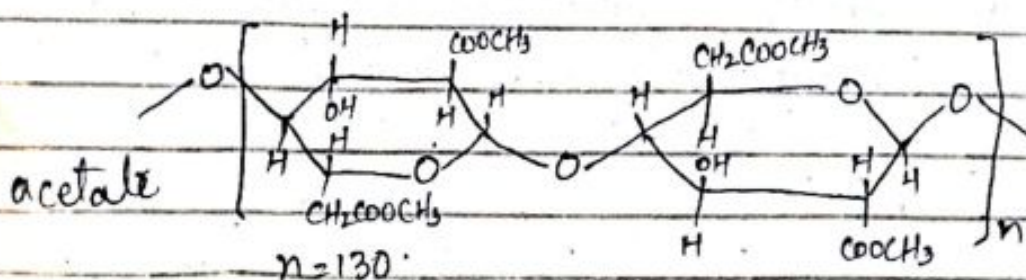
The swollen cellulose and a mixture of acetic acid (CH_3COOH) and anhydride is kept at a temperature of $25-30^\circ C$ for 8-10 hrs. to yield primary cellulose acetates or triacetate.

STEP #4:-

The primary cellulose acetate, i.e.; triacetate is hydrolysed or interacted with water (H_2O), such that only 2.3 to 2.4 acetyl or acetate groups occur per glucose unit.



Hydrolysis



⑨ Role of liming in chrome tanning.

In chrome tanning, liming plays a crucial role in initial stages of leather processing. Here's how it contribute to the process.

- Swelling & cleaning:- Liming involves soaking the raw material animal hides or skins in an alkaline soln. usually containing CaOH & sometimes NaOH . This step helps to swell & soften the hides, making it easier to remove hair, flesh and other unwanted materials.
- Hair Removal:-

The alkaline environment created by liming helps to break down the proteins in the hair, making it easier to mechanically remove the hair from hide.

- pH adjustment:-

Liming adjust the pH of the hides to a range where chrome tanning can be effectively carried out. Chrome tanning requires an acidic environment, and liming shifts the pH from acidic (in the case of fresh hides) to alkaline before the tanning process.

- Bacterial activity Control:-

Liming also has the added benefit of controlling bacterial growth during hide preservation.

⑩ Need of salting in manufacturing of leather.

Salting is an important step in the manufacture of leather for several reasons.

• Preservation:-

Salting helps to preserve the hides or skins by preventing bacterial growth and decomposition. It acts as a temporary preservative until further processing can take place.

• Removal of blood and extra moisture:-

Salting helps in removing blood and excess moisture from the hides or skins, which can lead to spoilage if not addressed. This prepares the material for subsequent processing.

• Dehydration:-

Salting aids in removing water from the hides or skins, which ~~is~~ can lead to spoilage is essential for proper preservation and prevent bacterial growth.

• Transport and storage:-

Salting allows for the safe transport and storage of raw hides or skins before they are processed further. It prevents them from deteriorating during transit.

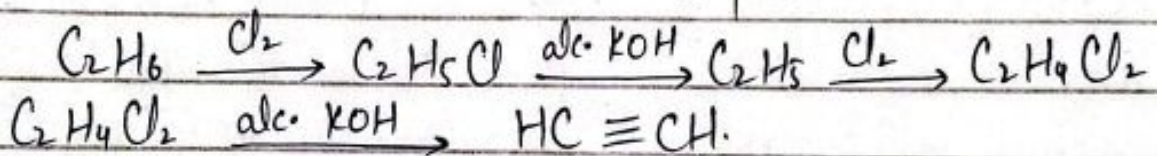
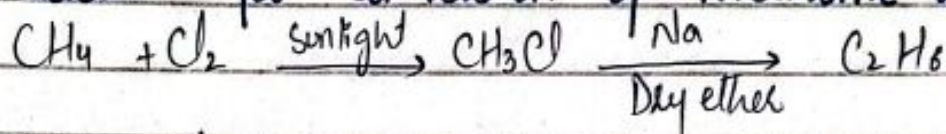
⑪ Enlist the industrial chem. produced from ethylene:-

Ethylene, a hydrocarbon gas, is a crucial feed-stock for the production of a wide range of industrial chemicals and products. Some of the key industrial chemicals produced from

ethylene:-

- | | |
|-----------------------------|--------------------------------|
| → Polyethylene (PE) | → Ethylene oxide |
| → Ethylene Glycols | → Polyvinyl alcohol (PVA) |
| → Polyvinyl chloride (PVC) | → Styrene |
| → Ethylene Dichloride (EDC) | → Vinyl chloride monomer (VCM) |
| → Alpha-olefin Derivatives | → Acetic acid. |

⑫ Reaction for conversion of methane to acetylene



⑬ Define nitration & write down possible products obtained from nitration of propane.

Nitration is a chemical process in which a nitro functional group ($-\text{NO}_2$) is introduced into a molecule or compound. It typically involves the substitution of hydrogen atom with $-\text{NO}_2$ through the reaction of the compound with a nitrating agent, HNO_3 , H_2SO_4 .

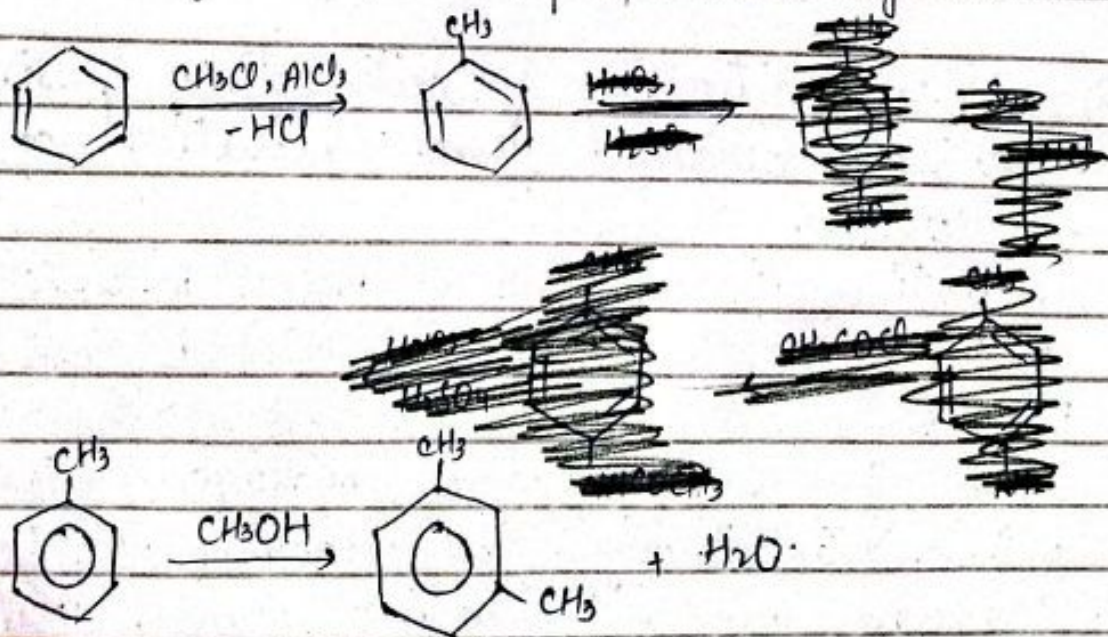
Possible products:- Nitro-alkane.

The specific products obtained and their yields would depend on the reaction conditions, including temperature, pressure and the presence of catalyst.

- 2-Nitropropane
- 1-Nitropropane
- Nitromethane
- Nitroethane.

⑭ Write down any one method of production of xylene. How it can be oxidized.

Xylene is produced by the methylation of Toluene and benzene. Commercial or laboratory grade xylene is produced usually contains about 40-65% of m-xylene & upto 20% each of o-xylene, p-xylene & ethyl benzene.



Xylenes are extensively oxidized to produce

important dicarboxylic acids, which are crucial monomers for the production of polyester. The specific product depends on xylene isomers being oxidized.

The oxidation of xylene involves the conversion of the methyl group on benzene ring to $-COOH$ groups. This is typically achieved via liquid phase catalytic oxidation with air to oxygen.

(15) Enlist the main tanning systems, also mentioned the most common system with reasons.

The main tanning systems includes:

- Vegetable tanning
- Mineral tanning
- Oil tanning
- Synthetic tanning
- Chrome tanning
- Aldehyde tanning
- Brain tanning
- Wet-White tanning

Most Common System:-

Chrome tanning is the most widely used methods and employs chromium salts, specifically chromium sulfate to tan the leather. It is known for its speed and effectiveness in producing soft and light coloured leather. Chrome tanned leather is used in many applications including upholstery, footwear & bags.

⑥ Describe the process during deliming & beating in leather industry.

Deliming:-

It is the process of removing the lime (CaOH) and other alkali used in the previous step, known as soaking or liming. It helps in preparing the leather for further process.

Beating:-

Beating is aimed at further softening the leather, improving its texture and removing any remaining non-collagenous proteins.

These processes are crucial for preparing the leather for dyeing, tanning and finishing stages. Deliming and beating helps to achieve the desired characteristics in the final leather product, such as softness, pliability and uniformity.

⑦ What are the major uses of C-black.

1. Pigments in Inks & paints:

C-black is used as a black pigment in inks, paints and coatings, providing a deep stable black color.

2. Inkjet & Toner Cartridges:

C-black is used in production of inkjet & toner cartridges for printers & photocopies.

3. Battery Electrodes:-

In the form of conductive carbon additives, carbon black is used in battery electrodes, improving electrical conductivity and performance in lithium-ion batteries.

4. UV Stabilization:-

In some applications, carbon black is used as a UV stabilizer to protect polymers and other materials from the damaging effects of UV radiation.

(18) Describe neutralization and dyeing in leather industry.

In leather industry, neutralization & dyeing are important processes that help to prepare and color leather.

Neutralizing:-

Neutralizing is a crucial step in leather processing, usually following the tanning process.

It involves adjusting the pH of the leather to neutral or slightly acidic conditions (pH 4.5-6.5) from the alkaline environment created during tanning.

The primary purpose of neutralizing is to stop the tanning reaction & stabilize the leather's structure.

This helps to prevent over-tanning, which can make the leather stiff and brittle. Proper neutralization ensures

that the leather is ready for subsequent processes like dyeing and finishing.

Dyeing :-

- It is the process of imparting color to leather. It is one of the most visually impactful stages in leather production, as it defines the final appearance of the leather product.
- Leather can be dyed using various methods including immersion dyeing, drum dyeing and spray dyeing, depending on the desired outcome and type of leather.
- Dyeing agents, such as synthetic or natural dyes are used to achieve the desired color. Dyeing also allows for the creation of different shades, tones and effects on leather, enhancing its aesthetic appeal.

(19) Cr salts are undesirable in leather industry. Comment.

Cr salts are generally undesirable in the leather industry due to their potential environmental health concerns. Cr is commonly used in the tanning process to convert animal hides into leather. There's two main forms of chromium used.

→ Trivalent chromium (Cr III)

This form of chromium is less toxic and is used in the tanning process to convert animal hides into leather. create high quality leather. Trivalent chromium is generally considered safe for both workers & the environment.

→ Hexavalent chromium (Cr VI)

This form of Cr is highly toxic & is by-product of certain tanning processes, particularly chrome tanning. It poses serious health risks to workers & can contaminate the environment if not properly managed. Exposure to hexavalent chromium lead to skin and respiratory issues, & its known as Carcinogen.

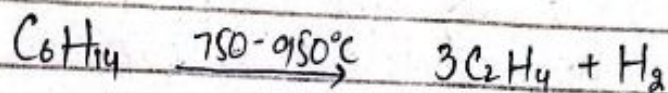
Q20 What are petrochemicals obtained from propene & acetylene.

Propene	Acetylene
Prop. Polypropylene (PP)	Vinyl chloride
Acetone	Acetaldehyde
Isopropanol	Ethylene
Propylene Oxide	Styrene
Isobutylene	Phenol.

21) Describe the process of production of ethylene from naphtha.

Ethylene is produced from naphtha, a petroleum based hydrocarbon mixture, through a process called steam cracking, also known as thermal hydrocarbon cracking.

Naphtha is heated to high temperatures (750-950°C) in the presence of steam within a furnace coil, causing the long hydrocarbon chains to break down into smaller molecules, including ethylene.



22) What are essential oil & veg. oil.

Essential oil

They are highly conc., volatile liq. extracted from various parts of aromatic plants (flowers, leaves, stems, bark, roots, fruits, seeds). They capture the essence of plant's scent and flavour, meaning the characteristic aromatic comp.

- lavender oil, peppermint oil, tea tree oil, lemon oil, eucalyptus, frankincense

Veg. oils

Veg. oils (carrier oils, fixed oils) are fats extracted from the seeds, fruits, nuts of various plants.

- composed of Triglycerides.
- Non-volatile, fatty, Nutritious, Extraction, Diverse use.

- Olive, coconut, sunflower, Canola, soybean, almond, jojoba, avocado oil.

②③ Name diff. beamhouse operations involved in manufacturing of leather.

These operations aim to clean the hides, remove unwanted materials (like hair, flesh and fats) and bring the collagen fibers into a suitable condition for tanning.

- Soaking
- Liming (Dehairing)
- Fleshing
- Splitting (optional)
- Deliming
- Bating
- Drenching (less common now)
- Pickling

②④ Write down different problems & diseases caused by tanning industries to human.

Here are different problems and diseases caused by tanning industry to human.

→ Respiratory problems:

- Chronic Bronchitis
- Occupational Asthma
- Irritation of the upper Respiratory Tract
- Pulmonary Edema, Bronchitis, Pneumonia
- Nasal Ulceration & Nasal Septal perforation
- Lung Cancer.

→ Skin Disorder:

- Eczema & allergic contact Dermatitis
- Skin Ulcerations
- Callosities

- chemical burns & rashes
- Leukoderma
- abdominal skin ulcers, scabies & other general skin diseases

→ Cancers:

- Lung cancer
- Bladder cancer
- Kidney cancer
- Pancreatic cancer
- Oral cavity cancer
- Nasal / Sinonasal cancer
- Soft tissue sarcoma
- Skin cancer & Melanoma
- Testicular cancer

→ Gastrointestinal Problem:

- Gastric Ulcers & Gastroenteritis
- Dysentery
- Diarrhea
- Jaundice
- Irritation of gastrointestinal mucosa
- Food chain contamination

→ Neurological effects:-

- Headaches & Dizziness
- Impairment of childhood learning & attention development
- Olfactory Function impairment
- Social memory Defects

→ Liver kidney Damage:

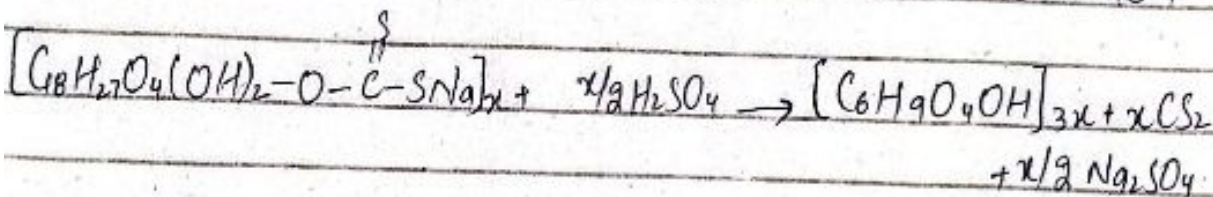
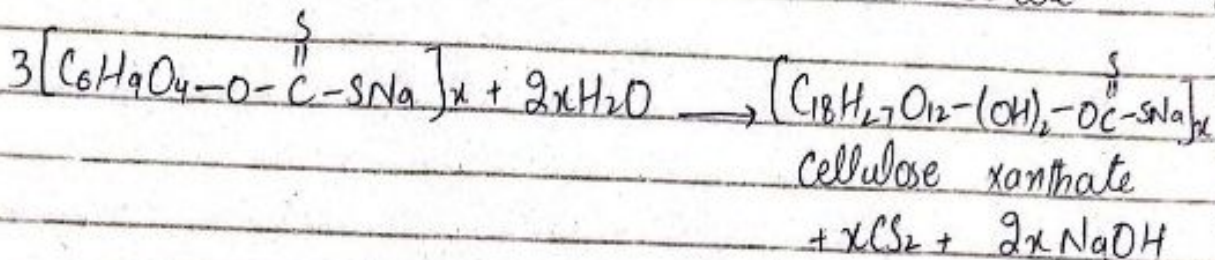
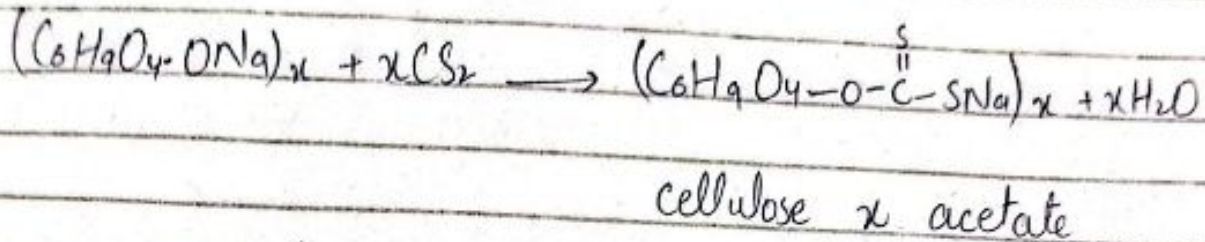
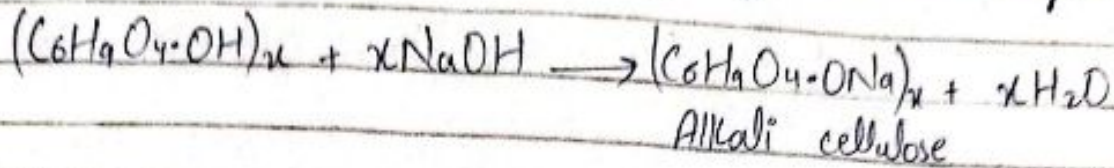
- Hepatotoxicity
- Kidney Inflammation

→ Other health issues:-

- Eye irritation & illness
- Hypertension
- Muscle fatigue
- Cardiovascular problem
- Poisoning to nervous system
- Genetic Defects
- Anemia
- Reproductive system prob.
- Anthrax
- Systemic toxicity

Textile Fibre

25) Write down chemical reactions involved in the manufacturing of viscose rayon



26) Describe mercerization & its importance in textile finishing.

Mercerizing:

Mercerizing is a textile finishing process, primarily applied to cotton, that enhance its properties. Cotton yarn or fabric is immersed under tension in a conc. soln. of NaOH at a specific temp. (15-20°C). The tension is crucial for achieving luster. After treatment, the

alkali is washed out and the fabric is neutralized with dilute acid.

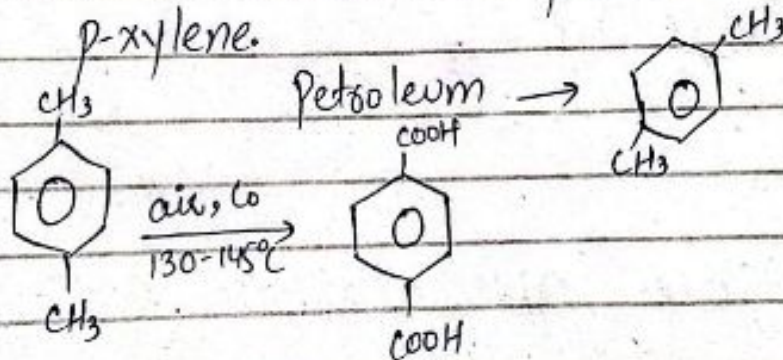
Benefits:

- Increase luster
- improve strength
- enhance dye uptake
- improved dimensional stability.

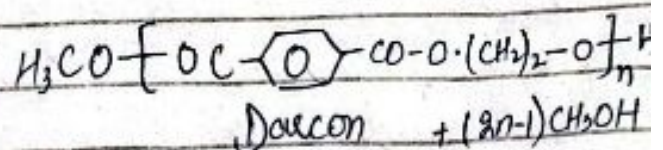
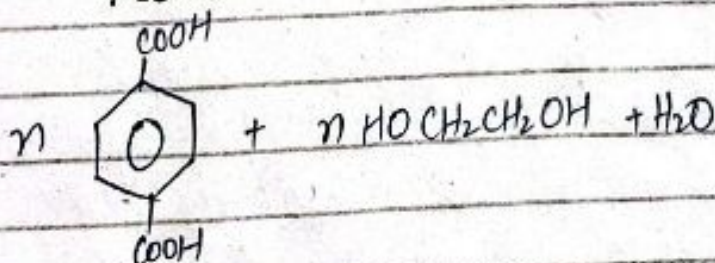
(27) Enlist the raw material for Dacron, also give the reaction of its formation.
Raw material:

Acid $\Rightarrow$ Terephthalic acid
Alcohol $\Rightarrow$ Ethylene glycol.

acid is obtained by the oxidation of p-xylene.



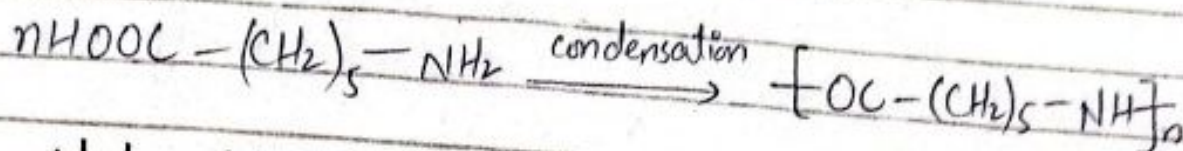
Reaction:



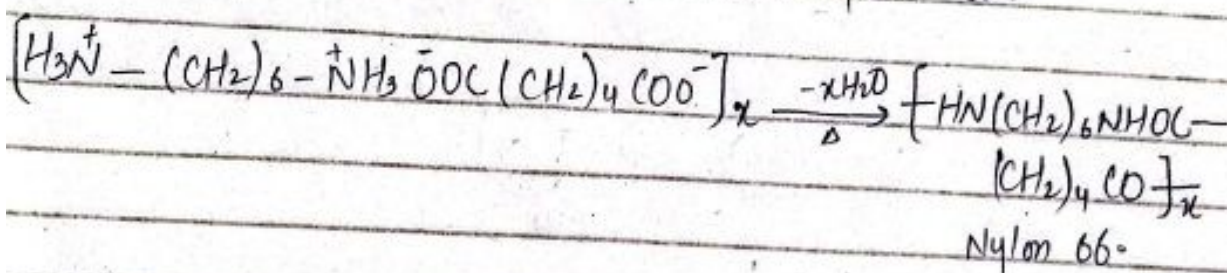
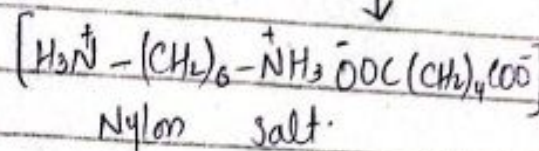
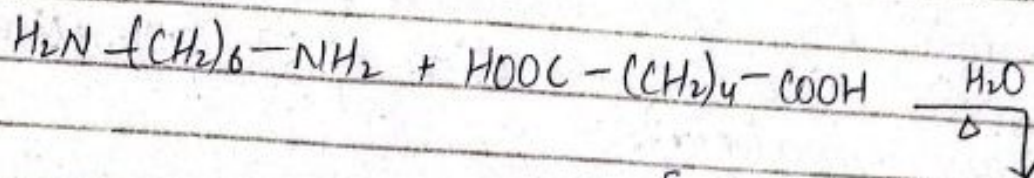
(28) Chemical eq. for production of Nylon 66 & Nylon 6.

Nylon 6:-

Nylon 6 is formed by the condensation of amino-caproic acid.



Nylon 66:-



(29) Define fibre. give attributes of fibre.

A fibre, in the context of materials & textiles, is a slender & elongated strand or filament composed of natural or synthetic substances. These filaments are typically long & thin, with a high length-to-diameter ratio. Fibers can be used to make a wide range of products.

including textiles, fabrics, ropes & composite materials.

There are 2 main categories of ^{fibres} fibres.

1. Natural fibres
2. Synthetic fibres

(30) Process of Singeing of cotton.

To burn off loose, protruding fibres (fuzz or hairiness) from the fabric surface. The cotton fabrics are quickly passed over a hot flame (usually gas flames) or heated plates/cylinders. The fabrics pass over brushes to raise the fibres then pass over a plate.

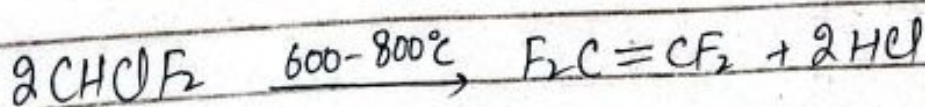
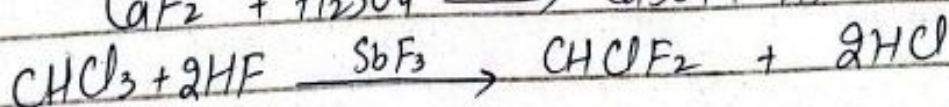
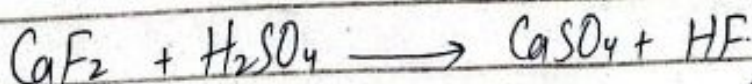
(31) Write down the raw material used for polyester & eq. for polyester.

Teflon:

Raw material:

~~CHCl₃~~ CHCl₃ and HF.

Eq:



(32) Name diff. processes used for the conversion of polymers to fibres.

- 1- Polymerization
- 2- ~~Cooling~~ Pellet formation
- 3- ~~Cutting & packing~~ Melting & extrusion
- 4- Cooling ~~Spinning~~ ~~cotton & cellulosic fibres~~
- 5- ~~Pellet~~ formation Drawing
- 6- ~~Drawing~~ Texturing
- 7- ~~Heat~~ Setting Cutting & Packing
- 8- ~~Twisting & plying~~ Heat Setting.
- 9- ~~Finishing~~
- 10- Spinning cotton & cellulosic fiber.
- 11- Twisting & plying
- 12- Weaving, knitting or Nonwoven formation.